Woochang Sim

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Research Interests

Adaptive Agents under Uncertainty | Bayesian & Meta-Reinforcement Learning | Probabilistic Modeling | Generalization to Unseen Environments

I study how intelligent agents can **represent uncertainty, infer hidden task and environment structure, and adapt their behavior through interaction**. I am currently exploring **belief-based approaches to generalization in unseen interactive environments**, with particular interest in Bayesian and meta-reinforcement learning methods for latent task inference, belief updating, and information gathering. Building on my previous work on abstract reasoning and ARC, I use **ARC-AGI** as a testbed for these questions and investigate how prior knowledge from pretrained models can support interactive adaptation. In the longer term, I am interested in distinguishing adaptation within a known hypothesis class from settings that require broader structural or model-class adaptation.

Education

2025 – Present **Ph.D. in AI Convergence**, Gwangju Institute of Science and Technology (GIST) Advisor: Prof. Sundong Kim

2023 – 2025 **M.S. in Artificial Intelligence**, Gwangju Institute of Science and Technology (GIST) Advisor: Prof. Sundong Kim

2016 – 2023 **B.S. in Computer Engineering**, Chosun University

Research Experience

Prior-Knowledge-Driven Generalization on ARC-AGI-3 *2026 – Present*

Ongoing Ph.D. Research

- Exploring how agents can **infer latent task and environment structure through interaction** when faced with environments that differ from those encountered during training.
- Investigating formulations based on **Bayes-Adaptive MDPs and meta-reinforcement learning**, with emphasis on belief updating, information gathering, and adaptation under task uncertainty.
- Exploring whether **prior knowledge from pretrained language models** can provide useful hypotheses about an environment and be refined using evidence gathered during interaction.
- Using **ARC-AGI-3** as an interactive testbed for studying prior-knowledge-driven adaptation and generalization to unseen tasks.

GIFARC: Synthetic Dataset for Leveraging Human-Intuitive Analogies to Elevate AI Reasoning *Apr 2025 – Present*

arXiv Preprint; Under Review

- **Introduced GIFARC**, an analogy-inspired synthetic dataset that transforms dynamic visual patterns from GIFs into executable ARC-style reasoning tasks paired with human-intuitive analogical information.
- **Developed a three-stage generation pipeline** that converts GIFs into structured visual abstractions, ARC-style task sketches, and executable task instances through vision-language and language models.
- **Evaluated how analogical information affects model reasoning** through controlled in-context learning and supervised fine-tuning experiments, including evaluation on downstream ARC benchmarks.
- **Led collaborative research execution**, coordinating task allocation, experimental planning, weekly research meetings, and consensus-building across collaborators.

Reasoning Abilities of Large Language Models: In-Depth Analysis on the Abstraction and Reasoning Corpus [Code] *Nov 2023 – Jan 2025*

Co-first Author, ACM Transactions on Intelligent Systems and Technology (TIST)

- **Operationalized a LoTH-based evaluation framework** into reproducible experimental settings for three reasoning properties: logical coherence, compositionality, and productivity.
- **Led the implementation of controlled reasoning experiments**, including prompting-based logical-coherence evaluations and an action-based compositionality environment with a custom interface for human baselines.
- **Compared state-of-the-art language models with human performance** across multiple abstraction and reasoning settings.
- **Identified substantial gaps between task performance and valid reasoning behavior**, highlighting limitations in current models' ability to perform robust abstract and compositional reasoning.

Publications

Lee, Seungpil*, **Sim, Woochang***, Donghyeon* Shin, Wongyu Seo, Jiwon Park, Seokki Lee, Sanha Hwang, Sejin Kim, and Sundong Kim (Jan. 2025). "Reasoning Abilities of Large Language Models: In-Depth Analysis on the Abstraction and Reasoning Corpus". In: *ACM Trans. Intell. Syst. Technol.* Just Accepted. ISSN: 2157-6904. DOI: 10.1145/3712701. URL: https://doi.org/10.1145/3712701.

Sim, Woochang*, Hyunseok* Ryu, Kyungmin* Choi, Sungwon Han, and Sundong Kim (May 2025). "GIFARC: Synthetic Dataset for Leveraging Human-Intuitive Analogies to Elevate AI Reasoning". In: *arXiv preprint arXiv:2505.20672*. Under review.

Lee, Jihwan*, **Sim, Woochang***, Sejin Kim, and Sundong Kim (2024). "Enhancing Analogical Reasoning in the Abstraction and Reasoning Corpus via Model-Based RL". In: *IJCAI Workshop on the Interactions between Analogical Reasoning and Machine Learning*.

Sim, Woochang, Hyebin Jin, Sejin Kim, and Sundong Kim (Feb. 2024). "The Possibility of Prompt Engineering for ARC Problem Solving". In: *The Korean Institute of Information Scientists and Engineers* 30.2, pp. 63–69. URL: https://www.kci.go.kr/kciportal/ci/sereArticleSearch/ciSereArtiView.kci?sereArticleSearchBean.artiId=ART003050661.

Gu, Gyojoon, **Sim, Woochang**, Jaegyun Im, Sejin Kim, and Sundong Kim (Dec. 2023). "Using Contrastive Learning for Abstraction and Reasoning Task". In: *Korea Software Congress*, pp. 828–830. URL: https://www.dbpia.co.kr/journal/articleDetail?nodeId=NODE11705255.

Seo, Wongyu, **Sim, Woochang**, and Sundong Kim (Dec. 2023). "Augmenting Few-Shot Demonstrations with Large Language Model". In: *Korea Software Congress*, pp. 341–343. URL: https://www.dbpia.co.kr/journal/articleDetail?nodeId=NODE11705093.

Technical Skills

Programming	Python
ML Frameworks & Tools	PyTorch, Hugging Face Transformers, Weights & Biases, Git
Research Methods	Experimental Design, Benchmark Construction, Ablation Studies, Statistical Evaluation, Human Evaluation, Inter-annotator Agreement Analysis
ML Experience	LLM Fine-tuning, In-context Learning, Synthetic Data Generation, Information Retrieval, Open-Domain Question Answering, Data Pipeline Development
Languages	Korean (Native), English (Intermediate)

Teaching & Service

Teaching Assistant, Machine Learning & Deep Learning, GIST *Sep – Dec 2025*

- Graded assignments on bias–variance tradeoffs, statistical learning theory, and core machine-learning concepts with detailed feedback.
- Taught fundamental deep-learning concepts covering neural networks, convolutional neural networks, and recurrent neural networks.

Student Representative, AI Convergence Department, GIST *Jan – Dec 2025*

- Organized department-wide student events, including a sports day and welcome reception for incoming graduate students.
- Facilitated communication between students and faculty members regarding department initiatives and student concerns.

Awards

Mar 2024 **The Chungshim Fellowship**
Jul 2023 **Best Presentation Award**, Korea Computer Congress 2023

Selected Projects

Open-Domain Question Answering System [Code] *Oct – Nov 2021*

Implemented a hybrid BM25/ElasticSearch/DPR retrieval pipeline with a RoBERTa-large reader over 60K Wikipedia documents; achieved **68.89 EM / 78.69 F1**, ranking 6th among 19 teams.

Relation Extraction with Typed Entity Markers [Code] *Sep – Oct 2021*

Implemented typed entity markers and domain-adaptive masked-language-model pretraining with RoBERTa-large, achieving a **72.6 F1 score**.

K-pop Relation Extraction Dataset Construction *Nov 2021*

Constructed a **2,468-instance** relation extraction dataset across 10 relation classes and developed iterative annotation guidelines, achieving **Fleiss'** $\kappa = 0.842$.